AI-Powered Agentic Portfolio Management System

Phase 2: Architecture, Prototype, and Evaluation Plan

Github: https://github.com/prithiraj-bhuyan/Portfolio-Agent-System/tree/main

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1. System Architecture

1.1 Architecture Overview

The system is organized into two layers, inspired by the TradingAgents framework (Xiao et al., 2025). **Layer 1 (Stock Analysis Engine)** processes a single ticker through a 7-stage pipeline: four concurrent analysts, a bull/bear researcher debate, a trader, a risk management team, a fund manager, and a human-in-the-loop gate. **Layer 2 (Portfolio Management)** orchestrates which tickers enter the pipeline and monitors portfolio-level concerns like concentration and rebalancing.

The pipeline is implemented as a **LangGraph StateGraph** with conditional edges that route the flow based on intermediate results. If the trader proposes HOLD, the pipeline short-circuits to the end node. If the risk team or fund manager rejects a proposal, the pipeline terminates early. If the human rejects a trade, execution is skipped. This design ensures no trade executes without passing through all required gates.

1.2 Pipeline Flow

Layer 1: Stock Analysis Engine (per ticker)

Stage	Agent(s)	Description
1	Analyst Team (4 agents)	Fundamental, Technical, Sentiment, and News analysts run concurrently. Each reads from tools (market data, indicators, social APIs, news feeds) and writes a structured AnalystReport to GlobalState.
2	Researcher Team (3 agents)	Bullish and Bearish researchers debate for 2 rounds using analyst reports as evidence. A Facilitator agent evaluates the debate and determines the prevailing view (BUY/HOLD/SELL) with a confidence score.
3	Trader Agent	Reads the debate outcome and analyst reports. Proposes a trade with action, quantity, price target, stop-loss, and rationale. Position sizing is capped at 20% of portfolio. If the view is HOLD, the pipeline terminates here.
4	Risk Team (3 perspectives)	Three risk perspectives (Aggressive, Neutral, Conservative) evaluate the proposal. Checks concentration limits, stop-loss requirements, cash sufficiency, and confidence thresholds. May adjust quantity or reject.
5	Fund Manager	Applies risk adjustments and makes the final approval decision. If risk-adjusted quantity is 0 or proposal was rejected, the pipeline terminates.

6	Human Gate	Every approved trade is surfaced for human review with full reasoning chain. The human can approve or reject. (Auto-approved in prototype; interactive in production.)
7	Execution	Executes the paper trade: updates cash, holdings, average cost basis, and logs the transaction. All state changes are atomic and traced.

Layer 2: Portfolio Strategist

- Selects tickers from watchlist for analysis
- Monitors portfolio concentration and rebalancing needs
- Triggers Layer 1 pipeline for each selected ticker

1.3 Conditional Edges and Stopping Conditions

The LangGraph pipeline includes three conditional routing points, each of which can short-circuit the pipeline:

1. **After Trader:** If action = HOLD, route to END (no point in risk-checking a non-trade).
2. **After Fund Manager:** If decision = HOLD (rejected), route to END.
3. **After Human Gate:** If human rejects, route to END. Otherwise, proceed to execution.

Additional stopping conditions: insufficient cash raises a ValueError caught by the execution node; API failures in analyst agents trigger rule-based fallbacks rather than pipeline failures.

1.4 Why This Architecture Over a Simpler Alternative

A simpler alternative would be a **single-agent approach**: one LLM prompt that takes in all data and outputs a trade decision. We chose the multi-agent architecture for three reasons:

1. **Specialization improves quality.** Each analyst focuses on one data domain (fundamentals vs. technicals vs. sentiment vs. news). A single prompt conflates these and produces shallower analysis, as shown empirically by the TradingAgents paper.

2. **Debate surfaces blind spots.** The bull/bear debate forces the system to consider both sides of a trade before committing. A single agent tends toward confirmation bias.

3. **Structured gates prevent costly errors.** The risk team and human-in-the-loop gates are distinct checkpoints. In a single-agent system, risk assessment and trade execution are bundled, making it harder to audit and override. Our architecture makes every decision point inspectable.

2. Role Definitions

Each agent has a defined role, input, output, and access pattern. The table below provides the complete specification:

Agent	Input	Output	Tools	LLM Usage
Fundamental Analyst	Ticker symbol	AnalystReport (signal, confidence, metrics)	MarketDataTool (fundamentals, price)	Groq llama-3.3-70b for analysis; rule-based fallback
Technical Analyst	Ticker symbol	AnalystReport (RSI, MACD, trend signal)	MarketDataTool (prices), TechnicalAnalysisTool	Groq for interpretation; rule-based fallback
Sentiment Analyst	Ticker symbol	AnalystReport (social score, volume)	SentimentTool (Reddit, X scores)	Rule-based only (no LLM needed for scoring)
News Analyst	Ticker symbol	AnalystReport (news sentiment, headlines)	NewsTool (headlines, macro context)	Rule-based scoring of article sentiments
Bullish Researcher	All AnalystReports for ticker	Argument JSON (bull case + key points)	Reads GlobalState	Groq LLM for argument generation
Bearish Researcher	All AnalystReports for ticker	Argument JSON (bear case + key points)	Reads GlobalState	Groq LLM for argument generation
Debate Facilitator	Debate rounds + reports	DebateRecord (prevailing view, confidence)	Reads GlobalState	Groq LLM or weighted-score rules
Trader	DebateRecord + reports + portfolio	TradeProposal (action, qty, targets)	MarketDataTool (current price)	Groq LLM for sizing; rule-based fallback

Risk Team (3 views)	TradeProposal + portfolio state	RiskAssessment (level, approved, adjustments)	MarketDataTool, PortfolioAnalyticsTool	Deterministic rules (no LLM)
Fund Manager	RiskAssessment + proposal	FinalDecision (action, qty, human flag)	Reads GlobalState	Deterministic logic (no LLM)
Human Gate	FinalDecision + reasoning chain	Approval boolean	None (interactive in production)	N/A
Portfolio Strategist	Watchlist + portfolio state	List of tickers to analyze	PortfolioAnalyticsTool, MarketDataTool	Deterministic rules

3. Coordination Logic

3.1 Who Starts

The Portfolio Strategist (Layer 2) initiates each cycle by selecting tickers from the watchlist. For each selected ticker, it invokes the LangGraph pipeline (Layer 1). In the prototype, all watchlist tickers are processed sequentially. In production, tickers could be prioritized based on price moves, news triggers, or rebalancing needs.

3.2 Handoff Mechanism

Agents communicate through **structured reports written to GlobalState**, not through natural language message passing. This follows the TradingAgents paper's structured communication protocol, which avoids the "telephone effect" where information degrades across many message exchanges. Each agent reads only the structured fields it needs from GlobalState, processes them, and writes its own structured report back. The LangGraph state (PipelineState) carries a reference to GlobalState and stage metadata.

3.3 Human-in-the-Loop

Every trade that passes through the fund manager's approval requires human sign-off before execution. In the prototype, this is auto-approved with a callback function. In the production dashboard, the system would surface the full reasoning chain (analyst reports, debate summary, risk assessment) and wait for the user's explicit approval or rejection. The user can also see proposed adjustments from the risk team and choose to accept or modify them.

3.4 How the Process Stops

The pipeline has four explicit termination points: (1) Trader proposes HOLD based on insufficient consensus. (2) Risk team rejects the trade (e.g., insufficient cash, concentration too

high). (3) Fund manager reduces quantity to zero due to risk adjustments. (4) Human rejects the trade. Each termination is logged in the interaction trace with the reason.

4. Tools, Memory, Data, and State Design

4.1 Tools

Tool	Data Source	Access Pattern
MarketDataTool	yfinance API (mock fallback with realistic per-ticker data)	Read-only. Called by Fundamental Analyst, Technical Analyst, Trader, Risk Team.
TechnicalAnalysisTool	Computed from price history (RSI, MACD, Bollinger, SMA, trend)	Read-only. Called by Technical Analyst.
SentimentTool	Mock social sentiment (Reddit, X/Twitter scores). Production: FinBERT + API.	Read-only. Called by Sentiment Analyst.
NewsTool	Curated news headlines with sentiment scores. Production: NewsAPI or Bloomberg.	Read-only. Called by News Analyst.
PortfolioAnalyticsTool	Computed from portfolio state (concentration, P&L, weights)	Read-only. Called by Risk Team, Portfolio Strategist.

4.2 Memory and State

The system uses a **GlobalState object** as the single source of truth, shared across all agents within a cycle. This is an in-memory state store (not a database) that holds:

- **PortfolioState:** Cash, holdings (ticker → shares), average cost basis per position, total portfolio value.
- **Analyst Reports:** Dictionary of ticker → list of AnalystReport objects. Each report has signal, confidence, metrics, reasoning.
- **Debate Records:** Dictionary of ticker → DebateRecord (rounds, prevailing view, facilitator summary).
- **Trade Proposals, Risk Assessments, Final Decisions:** One per ticker, written sequentially as the pipeline progresses.
- **Transaction Log:** Append-only list of executed trades.
- **Interaction Trace:** Full audit trail of every agent action with timestamp, agent name, action type, and details.

Each agent writes to GlobalState through type-safe setter methods (e.g., add_analyst_report, set_proposal) that also append to the interaction trace. This ensures every state mutation is logged. Agents read from GlobalState using dictionary lookups by ticker. This design prevents agents from accidentally modifying each other's outputs.

4.3 Why These Decisions Make Sense

Structured reports over natural language messages: Following the TradingAgents paper, we use typed dataclasses (AnalystReport, DebateRecord, etc.) rather than free-text messages. This prevents information loss across the pipeline and makes each handoff inspectable.

In-memory state over database: For a paper-trading prototype, an in-memory store is sufficient and avoids infrastructure complexity. The GlobalState is serializable to JSON for persistence between sessions in a production version.

Append-only interaction trace: This supports the course requirement for evidence logs and makes it easy to debug failures by replaying the exact sequence of agent actions.

5. Prototype and Interaction Trace

5.1 Prototype Overview

The prototype is a fully functional Python application built with LangGraph and Groq (llama-3.3-70b-versatile, free tier). It runs end-to-end: from ticker selection to trade execution, producing structured logs at every stage. The system runs with deterministic mock data when API keys are unavailable, making it reproducible for evaluation.

5.2 Technology Stack

- **Orchestration:** LangGraph StateGraph with conditional edges
- **LLM:** Groq API (llama-3.3-70b-versatile, free tier) with rule-based fallbacks
- **Market Data:** yfinance (live) or deterministic mock data
- **Language:** Python 3.11+
- **State Management:** Custom GlobalState with typed dataclass protocol

5.3 Sample Interaction Trace

Below is the trace from a prototype run analyzing AAPL, GOOGL, and MSFT with $100,000 starting cash:

AAPL Pipeline:

Agent	Signal	Confidence	Key Detail
Fundamental	HOLD	0.59	PE=28.5 (high), but strong ROE=1.609 and margins=26.4%
Technical	HOLD	0.40	RSI neutral, MACD bullish, trend sideways
Sentiment	HOLD	0.40	Mixed social sentiment, score near neutral

News	BUY	0.58	Avg sentiment +0.413: strong earnings, AI features offset China concern
Debate	HOLD	0.44	Mixed signals insufficient for consensus; facilitator rules HOLD
Trader	HOLD	0.44	No trade proposed due to HOLD consensus

Outcome: HOLD - pipeline short-circuited after trader stage. No trade executed.

GOOGL Pipeline:

Agent	Signal	Conf	Key Detail
Fundamental	BUY	0.75	PE=22.3, growth=13.9%, low debt (0.42), strong ROE
Technical	BUY	0.52	MACD bullish, uptrend, RSI neutral
Sentiment	BUY	0.60	Positive social sentiment, score=0.445
News	HOLD	0.54	Cloud growth and AI positive, antitrust is concern
Debate	BUY	0.54	3/4 analysts bullish; bull case wins debate
Trader	BUY 65	0.54	Allocating ~12% of $100K cash
Risk Team	LOW	Approved	Concentration 12.0% < 20% limit, stop-loss set
Fund Manager	BUY 65	Approved	No adjustments needed
Human Gate	—	Approved	Auto-approved (prototype mode)
Execution	BUY 65	$11,965.20	GOOGL @ $184.08. Cash after: $88,034.80

Outcome: EXECUTED - full pipeline completed. 65 shares of GOOGL purchased.

MSFT Pipeline:

Agent	Signal	Conf	Key Detail
Fundamental	BUY	0.75	Growth=16.2%, margin=35.6%, ROE=38.9% (outstanding)

Technical	HOLD	0.40	MACD bullish but RSI elevated, trend sideways
Sentiment	BUY	0.51	Positive social sentiment
News	BUY	0.58	Azure 31% growth, Copilot adoption driving revenue
Debate	BUY	0.54	3/4 bullish; strong fundamentals prevail
Trader	BUY 21	0.54	Allocating ~12% of remaining $88K cash
Risk Team	LOW	Approved	Concentration 11.5% < 20% limit
Fund Manager	BUY 21	Approved	No adjustments needed
Execution	BUY 21	$10,177.02	MSFT @ $484.62. Cash after: $77,857.78

Outcome: EXECUTED - 21 shares of MSFT purchased.

Final Portfolio: $77,857.78 cash + 65 GOOGL ($184.08) + 21 MSFT ($484.62) = $99,999.00. Two trades executed, one ticker held. AAPL was correctly filtered out due to insufficient consensus.

6. Evaluation Plan

We define seven test scenarios covering happy paths, coordination logic, risk management, error handling, portfolio management, observability, and existing position handling. Each test has explicit success criteria and measures.

#	Scenario	Category	Success Criteria	Measure	Status
1	**Strong consensus BUY**	End-to-end	When 3+ analysts signal BUY, a trade executes and holdings update	Trade executed, holdings non-empty	**PASS**
2	**Mixed signals → HOLD**	Coordination	When analysts disagree, debate resolves to HOLD and no trade is proposed	Outcome contains HOLD	**PASS**
3	**Concentration limit**	Risk mgmt	Single position never exceeds 20% of portfolio value	Concentration ≤ 20% (1% tolerance)	**PASS**

4	**Insufficient cash**	Error handling	Trade blocked when portfolio has <$500 and stock costs >$400	No crash, cash ≥ 0, trade blocked	**PASS**
5	**Multi-ticker portfolio**	Portfolio mgmt	System handles 3 tickers, builds diversified portfolio without over-allocating	≥1 position, cash > 0	**PASS**
6	**Trace completeness**	Observability	Interaction trace includes entries from all agent types	All 5+ agent types present in trace	**PASS**
7	**Existing position handling**	End-to-end	Pipeline completes gracefully when stock is already held	No crash, correct outcome	**PASS**

6.1 Planned Phase 3 Test Additions

For Phase 3, we plan to expand the evaluation with:

- **Backtesting over historical periods:** Run the system over 3-month windows of actual market data and measure cumulative return, Sharpe ratio, and maximum drawdown vs. buy-and-hold.
- **Adversarial scenarios:** Flash crash simulation, conflicting breaking news, and API failure cascades.
- **LLM vs. rule-based comparison:** Run identical scenarios with and without Groq LLM to measure quality improvement from LLM reasoning.
- **Latency and cost profiling:** Measure per-ticker pipeline latency and Groq API token costs.

7. Risk and Governance Plan

7.1 Identified Risks and Mitigations

Risk	Likelihood	Impact	Mitigation

LLM hallucination produces invalid trade signal	Medium	A fabricated metric could lead to a bad trade recommendation	Every LLM output is parsed and validated against the Signal/TradeAction enum. Invalid JSON falls back to rule-based logic. Risk team provides a second check.
Groq API rate limit or downtime	High	Pipeline stalls if LLM calls fail	All agents have deterministic rule-based fallbacks. Pipeline degrades gracefully to rules-only mode without crashing.
Concentration risk - over-allocation to one stock	Medium	Single stock crash wipes out portfolio	Hard 20% cap enforced by Risk Team. Fund Manager zeros out any trade that would exceed the limit.
Stale or delayed market data	Medium	Decisions based on outdated prices	Paper trading mitigates real financial loss. Production would use real-time feeds with staleness checks.
Scope creep - 13 agents too complex for course timeline	Medium	Incomplete implementation	Core 7-stage pipeline is functional now. Remaining agents (if any) can be stubbed with rule-based logic.
Human override fatigue	Low	User auto-approves everything, defeating the safety gate	Dashboard will show a risk score and highlight high-risk trades visually. Future: require explicit rejection reason.
Model bias toward bullish signals	Medium	System over-buys in a bear market	Bear researcher explicitly argues against; facilitator weighs both sides. Backtesting against bear markets planned for Phase 3.

7.2 Governance Controls

- **Human-in-the-loop:** Every trade requires explicit human approval before execution. No autonomous trades.

- **Full audit trail:** Every agent action is logged in the interaction trace with timestamp, agent identity, and details. Traces are saved as JSON and can be replayed for debugging.

- **Paper trading only:** The system executes simulated trades only. No real money is at risk. This is a deliberate scope boundary.
- **Position limits:** Hard cap of 20% concentration per position. Enforced at the risk team level, not just the trader level, preventing bypass.
- **Graceful degradation:** If the LLM API is unavailable, all agents fall back to deterministic rules rather than failing. This prevents the system from being blocked by external dependencies.

7.3 Trust and Misuse Concerns

The system is designed for educational and paper-trading purposes. If deployed in a real-money context, additional safeguards would be needed: daily loss limits, circuit breakers, regulatory compliance checks, and a review process for the LLM prompts used by each agent. We also acknowledge that LLM-generated financial analysis should not be treated as professional investment advice.

8. Contribution Update

Below is the team contribution breakdown for Phase 2:

Team Member	Contribution	Hours	Status
Raj Bhuyan	Architecture design, LangGraph pipeline, orchestrator	~12 hrs	Complete
Sanath Mahesh Kumar	Agent implementations (analysts + researcher team)	~12 hrs	Complete
Mahir Nagersheth	Tools, evaluation plan, test scenarios	~10 hrs	Complete